$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^\dagger & -\frac{Y_1 k^2}{6} & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & Y_2 k^2 + 3\,\overset{(2)}{\kappa}_{1}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & Y_3 k^2 + 2\,\overset{(2)}{\kappa}_{1} & \frac{Y_3 k^2 + 2\,\overset{(2)}{\kappa}_{1}}{\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_3 k^2 + 2\,\overset{(2)}{\kappa}_{1}}{\sqrt{2}} & \frac{1}{2}\left(Y_3 k^2 + 2\,\overset{(2)}{\kappa}_{1}\right) & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{Y_4 k^2}{18} & -\frac{Y_4 k^2}{9\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{Y_4 k^2}{9\sqrt{2}} & -\frac{Y_4 k^2}{9}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{Y_4 k^2}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^\dagger & \frac{1}{\mathrm{Det}(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(0^-)}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3\,\mathrm{Det}(1^+)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{1}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{1}{3\,\mathrm{Det}(1^-)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{2}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(2^-)}
\end{array}
$$

Abbreviations used in matrices

$$Y_1 == 9\,\overset{(4)}{\kappa}_{1} + 5\,(2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16})\ \&\&\ Y_2 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16}\ \&\&\ Y_3 == 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3}\ \&\&$$

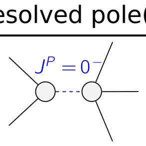
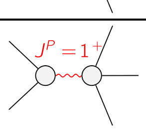
$$Y_4 == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}\ \&\&\ \mathrm{Det}(0^+) == -\frac{1}{6} k^2\,(9\,\overset{(4)}{\kappa}_{1} + 5\,(2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}))\ \&\&$$

$$\mathrm{Det}(1^+) == k^2\,(3\,\overset{(4)}{\kappa}_{15} - \frac{3\,\overset{(4)}{\kappa}_{3}}{2}) + 3\,\overset{(2)}{\kappa}_{1}\ \&\&\ \mathrm{Det}(2^-) == \frac{1}{2} k^2\,(2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16})$$

Lagrangian

$$
\begin{aligned}
&\overset{(2)}{\kappa}_{1}\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(2)}{\kappa}_{1}\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_{1}\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \\
&\frac{10}{9}\,\overset{(4)}{\kappa}_{15}\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \frac{5}{9}\,\overset{(4)}{\kappa}_{16}\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{3}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \\
&4\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + 2\,\overset{(4)}{\kappa}_{16}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 2\,\overset{(4)}{\kappa}_{3}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \\
&2\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \overset{(4)}{\kappa}_{16}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_{3}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{8}{3}\,\overset{(4)}{\kappa}_{15}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \\
&\frac{4}{3}\,\overset{(4)}{\kappa}_{16}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{1}{2}\,\overset{(4)}{\kappa}_{3}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == \mathcal{J}^{\#2\,\alpha\beta}_{1^+}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	11	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$\frac{3\,\overset{(2)}{\kappa}_{1}}{-\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$	$\frac{1}{-\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$
	3	$\frac{2\,\overset{(2)}{\kappa}_{1}}{-2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{3}}$	$\frac{2}{6\,\overset{(4)}{\kappa}_{15} - 3\,\overset{(4)}{\kappa}_{3}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	